

◆ AGENTIC URBAN INTELLIGENCE PLATFORM · 2026

Urban*Pulse* AI

—
AI-powered civic issue reporting, intelligent routing,
automated prioritization & real-time city analytics.

AI AGENTS

9

LangGraph DAG

TRIAGE TIME

12_s

Report → Officer

Civic complaints vanish into black holes.

Every day, municipal corporations receive thousands of reports on potholes, water leaks, garbage, and streetlight failures — routed manually, prioritized by whoever shouts loudest, and resolved without accountability.



FOR CITIZENS

No channel. No status. No trust.

Phone calls, WhatsApp, in-person visits — reports enter a void with no acknowledgement or ETA.



FOR OFFICERS

Manual triage. No priority.

Every complaint arrives with equal urgency. Hours lost on triage that should be spent on repairs.



FOR ADMINS

Zero city-wide visibility.

No health score, no trending issues, no data-driven resource allocation. Only headlines.



EXISTING SYSTEMS

Glorified CRUD forms.

FixMyStreet-style apps are complaint boxes — no AI, no routing, no analytics, no accountability.

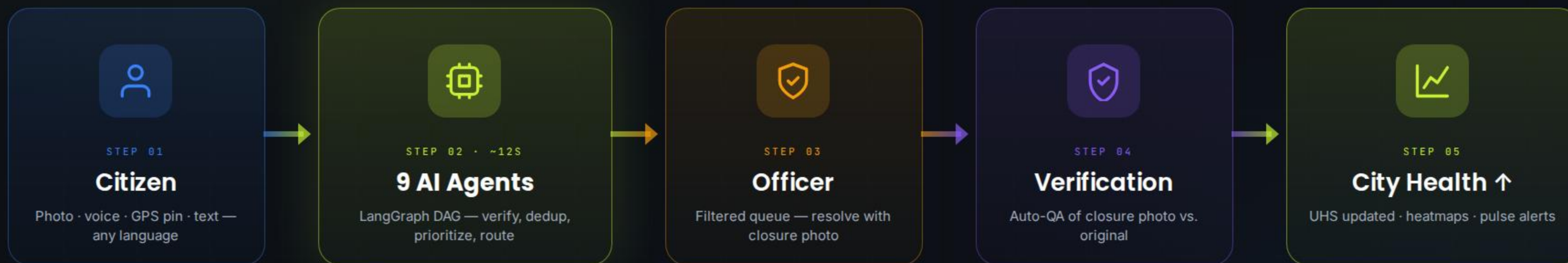
Not another complaint app.

A triage platform.

CAPABILITY	EXISTING SOLUTIONS FixMyStreet · IChangeMyCity · WhatsApp bots	◆ URBANPULSE AI Agentic triage & routing platform
Report routing	✗ Manual dispatcher	✓ AI routing agent · category → department rules
Prioritization	✗ First-come, first-served	✓ Severity scoring · category + location + text NLP
Duplicate detection	✗ None — every submission a new ticket	✓ PostGIS 50m radius + semantic embedding match
Live transparency	✗ Black-box — no status updates	✓ Real-time SSE agent trace visible to citizens
Analytics & scoring	✗ Basic ticket counts	✓ Urban Health Score (0–100) · pulse alerts · UHS/ward
AI architecture	✗ None — keyword filters at most	✓ 9-agent LangGraph DAG · Gemini vision + reasoning

◆ THE SHIFT From a *complaint box* to an **autonomous triage engine** — the same jump CRMs made a decade ago.

*One report. **Nine agents.**
Zero black boxes.*



◆ MULTI-MODAL INPUT

Photo · voice · GPS · text in any language
— CX agent normalizes it.

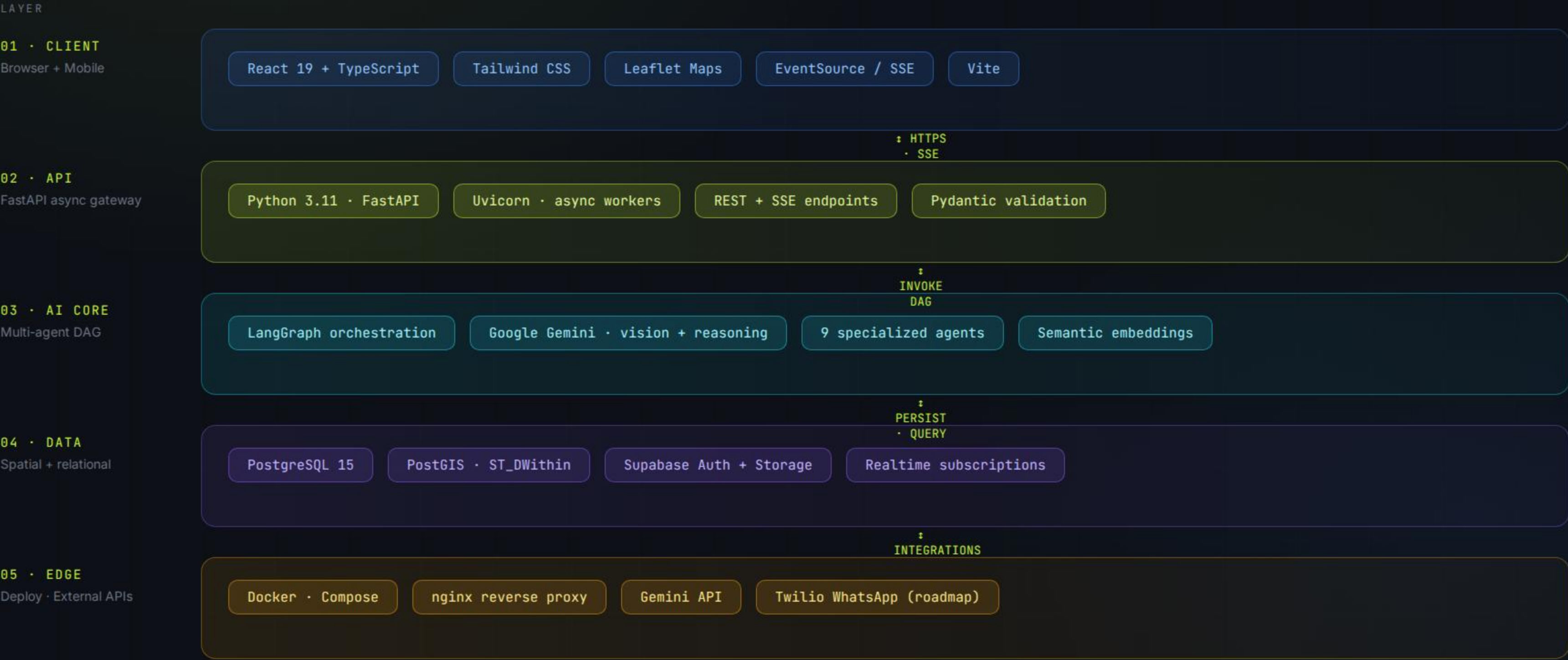
◆ ROLE-BASED DASHBOARDS

Citizen · Officer · Dept Head · Admin ·
Super Admin — six curated views.

◆ LIVE SSE TRACE

Every AI decision streamed in real time —
no black boxes for anyone.

Five layers. Built to scale.



Nine agents. One directed graph.

Independent nodes run in parallel; a Priority + Routing chain gates the middle; three finalization agents fan out. Every step streams its reasoning over Server-Sent Events.

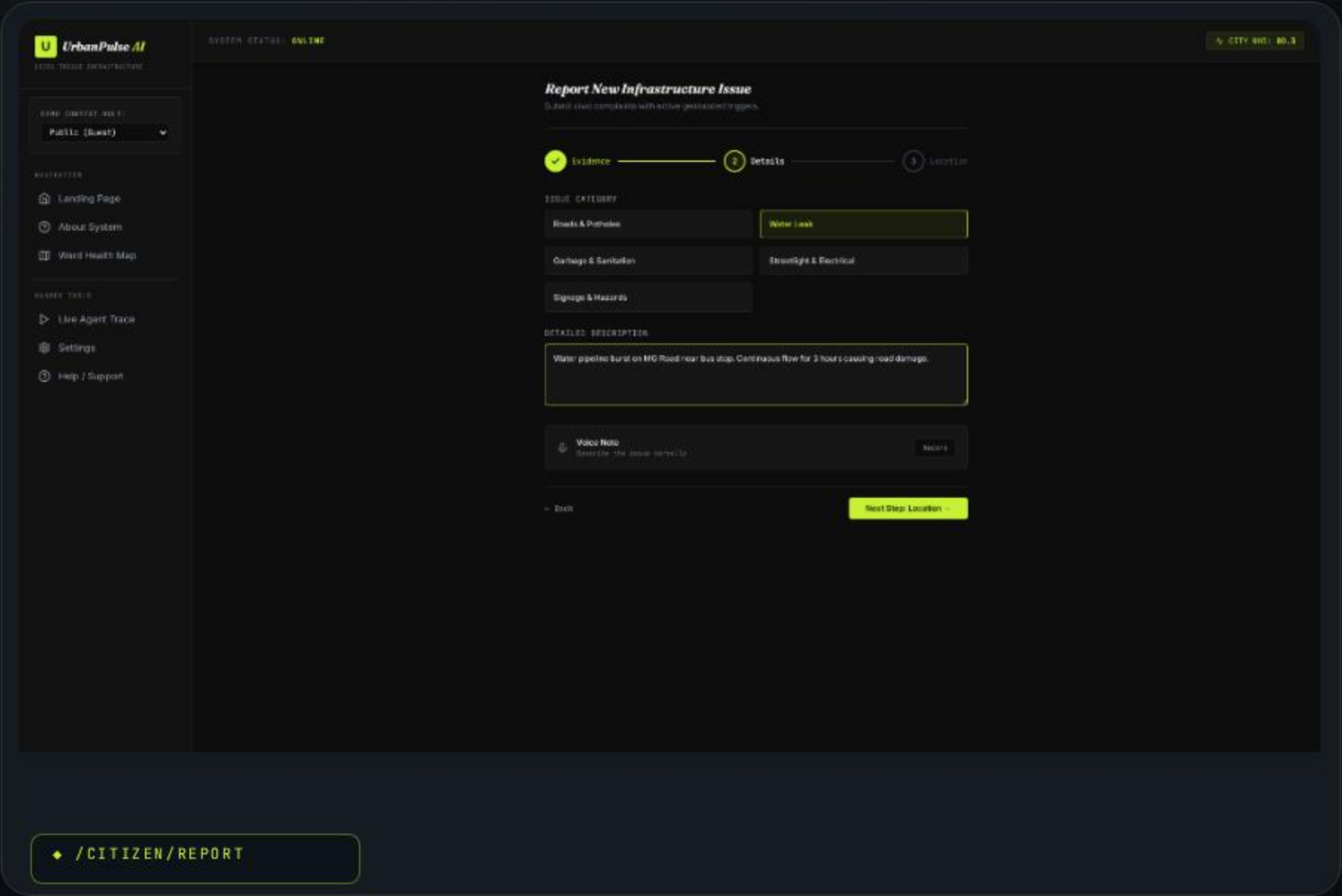
● ~12S END-TO-END



◆ LIVE SSE STREAM · AGENT TRACE VISIBLE TO CITIZENS

→ 00:02 CX Agent: language=en, intent=infrastructure_report | → 00:04 Vision: damage=severe, category=water_leak | → 00:06 Dedup: 0 duplicates in 50m radius
→ 00:08 Priority: severity=3 (high) | → 00:10 Routing: dept=Water · officer=28dca4a5 | → 00:12 Analytics: UHS Ward-1 → 74.5 (-2.0)

Report an issue in *under a minute.*



CITIZEN VALUE

No account needed. No forms. Photo, voice in any language, GPS pin — submitted once, tracked forever via live SSE trace + real-time status.

01

EVIDENCE

Snap a photo or record a voice note

Optional — the Vision Agent runs damage assessment; CX Agent transcribes and translates any language.

02

DETAILS

Pick a category · describe briefly

Roads · Water Leak · Garbage · Streetlight · Signage. Text is optional — voice + photo alone are enough.

03

LOCATION

Drop a pin · submit & watch AI think

Draggable GPS marker. Then a live SSE stream shows 9 agents processing your report in real time.

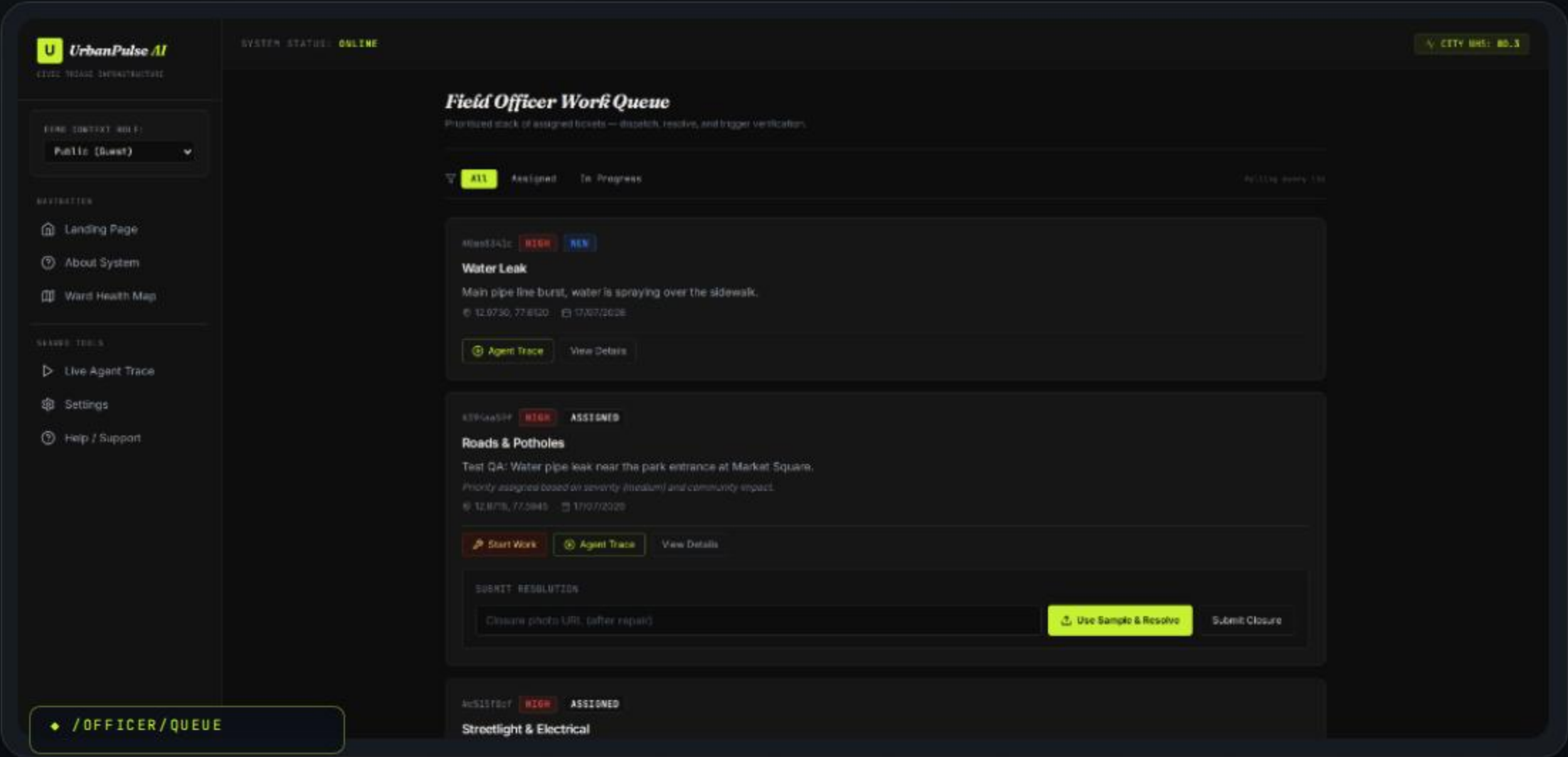
Field officers act *on triaged intelligence*.

No more inbox chaos. Officers open the app to a pre-sorted queue — priority-ranked, categorized, geo-located, and pre-routed. One-click resolve with closure photo triggers automatic AI verification.

TIME SAVED PER TICKET

~15min

vs. manual triage baseline



01 · PRIORITIZED

HIGH · MEDIUM · LOW badges auto-assigned
Severity from Priority Agent · sorted top-of-queue

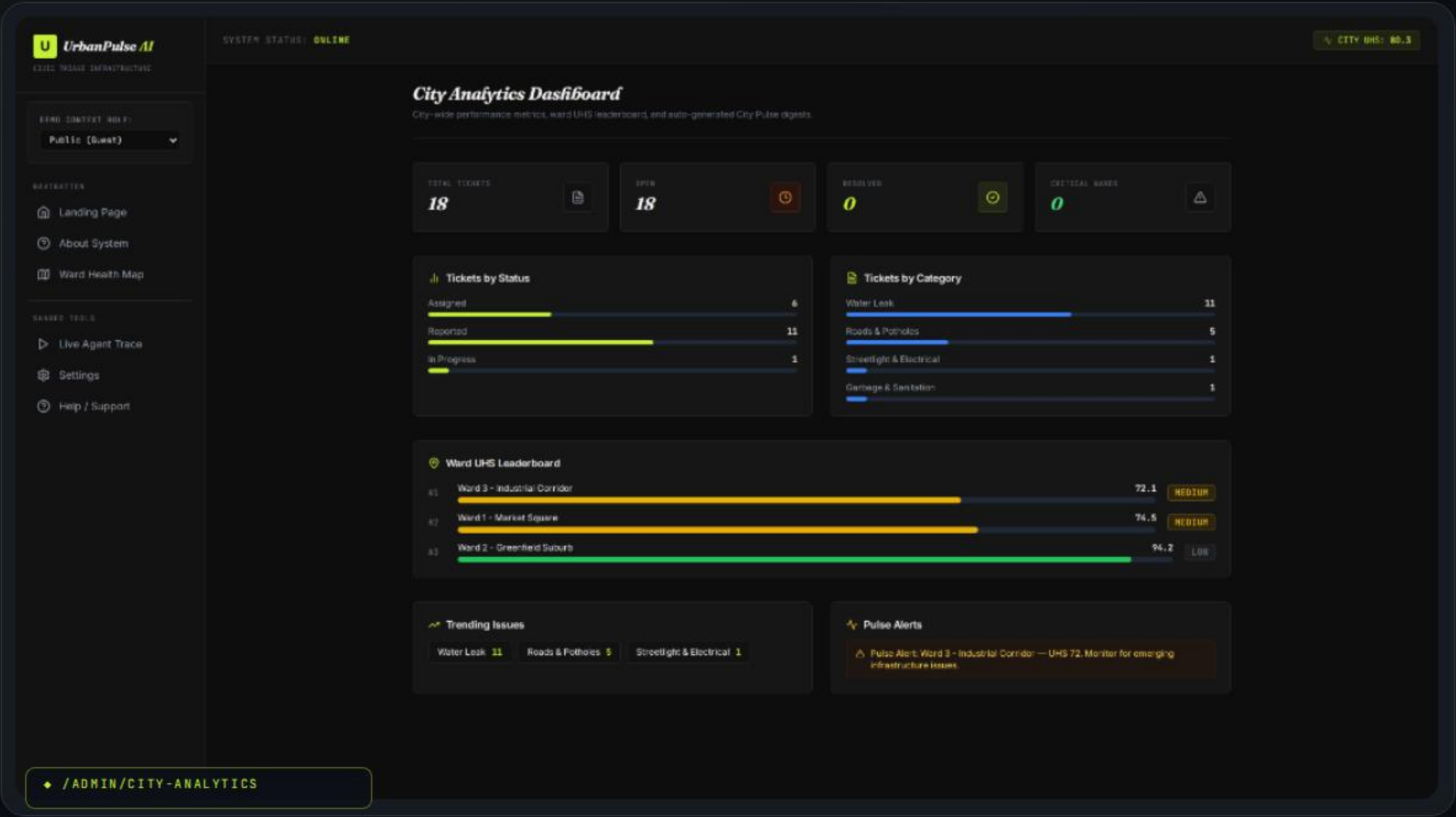
02 · GEO-LOCATED

GPS coords + one-tap directions
Every ticket carries lat/long + Agent Trace link

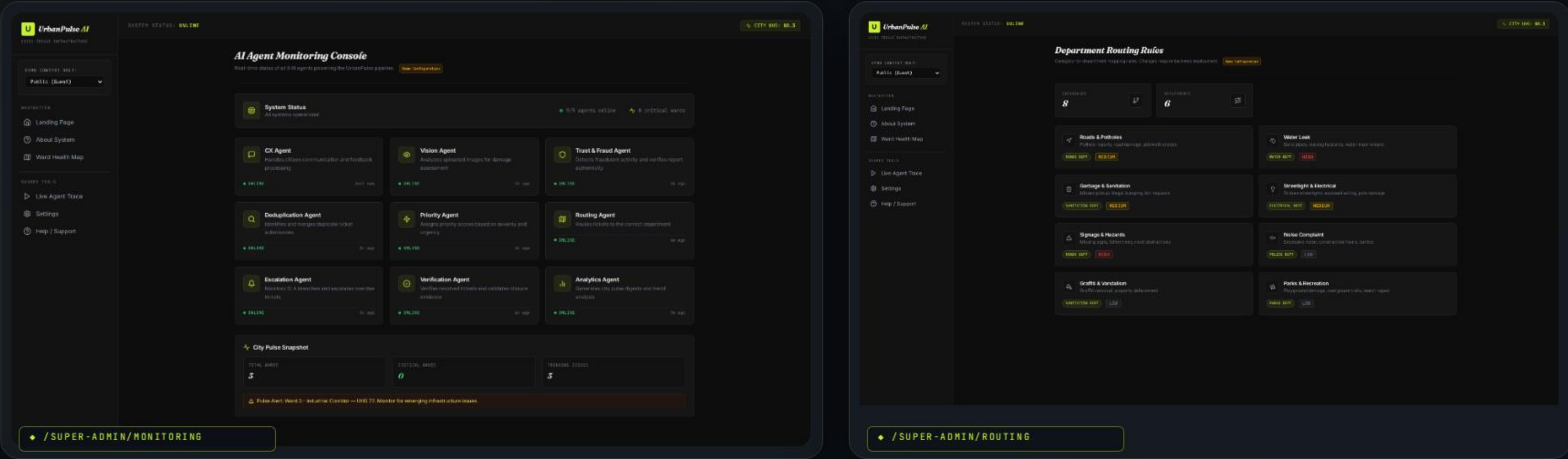
03 · CLOSURE PHOTO

Upload → Verification Agent auto-approves
Gemini matches repair vs. original — no paper QA

City-wide intelligence, *live*.



Full-stack oversight for *system operators*.



AGENT HEALTH

9/9 agents online · latency, error rate, last-seen per node

ROUTING RULES

8 categories → 6 departments · editable priority defaults

USER MANAGEMENT

7 roles · Supabase RLS · department assignment

AUDIT LOG · IMMUTABLE

Every admin action logged · JSONB detail · compliance-ready

A water leak, from 8:15 AM to closure.

A single ticket traced through every role, every agent — the same demo we walk judges through.

T + 00:00

Citizen submits

Photo + voice + GPS pin at MG Road.
Category: Water Leak.

T + 00:14

Officer notified

Ticket appears atop nearest officer's
queue with 4h SLA.

T + 02:32

Auto-verified

Verification Agent matches
before/after photos — ticket resolved.



T + 00:12

AI pipeline done

9 agents complete. Severity: HIGH.
Routed to Water Dept.

T + 02:30

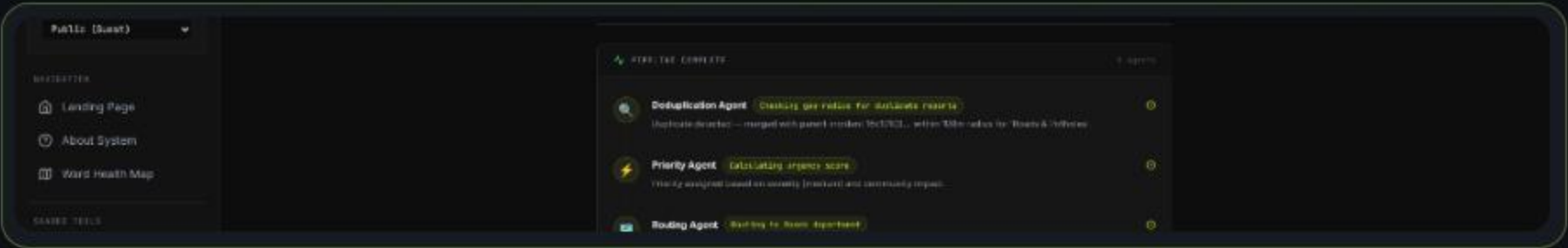
Repair on site

Officer starts work, uploads closure
photo of sealed pipe.

T + 02:33

UHS updated

Analytics Agent recomputes Ward-1
UHS · pulse alert cleared.



◆ TOTAL CITIZEN-TO-CLOSURE

~2h 33m — every step visible, every AI
decision streamed, every stakeholder in the
loop.

Why judges should look twice.

01

Multi-Agent DAG orchestration

LangGraph directed acyclic graph — 9 specialized agents run in parallel where independent, gated where sequential. Not a chained prompt.

02

Live SSE agent trace

Every agent's reasoning streamed to the browser via Server-Sent Events. Explainable AI you can watch — no black box for citizens or judges.

03

Trust & fraud scoring

Reputation model + spam signals + missing-media checks catch bad-faith reports before they clog the queue.

04

Spatial + semantic dedup

PostGIS ST_DWithin (50m) + embedding similarity — the same pothole reported 12 times becomes one ticket.

05

Explainable priority

Severity score is not a black-box number — Priority Agent surfaces the reasons (proximity to school/hospital, hazard class, description NLP).

06

Urban Health Score

Composite 0–100 metric per ward — proprietary weighting of resolution time, density, severity, SLA compliance. Actionable, not vanity.

07

SLA-first escalation

Every ticket carries a severity-based deadline. Escalation Agent triggers pulse alerts before public outrage, not after.

08

Docker-first horizontal scale

Containerized FastAPI + async workers + PostGIS scale-out. Ready for 10K reports/day with Celery + Redis when growth demands.

From ward-level triage to national digital twin.

◆ NOW • SHIPPED

Foundations.

Everything the demo just showed.

- 9-agent LangGraph pipeline
- Role-based dashboards (7 personas)
- Live SSE agent trace
- PostGIS spatial dedup
- Urban Health Score analytics
- Docker-ready deployment

▶ NEXT • Q3 / Q4 2026

Sensors.

Physical world feeds the agent pipeline directly.

- IoT sensor ingestion (water, air, noise)
- CCTV pothole & garbage detection
- Drone closure-photo verification
- Twilio WhatsApp report intake
- React Native mobile officer app
- Production auth & rate limiting

▶▶ FUTURE • 2027+

Predict.

The city stops reacting — it forecasts.

- Predictive failure analytics
- Flood & drainage forecasting
- Traffic congestion optimization
- City-scale digital twin
- Fine-tuned municipal LLMs
- National smart-city rollout

*Built with what the **best teams** ship on.*



FRONTEND

React 19 · TypeScript

- + Tailwind CSS
- + Vite build
- + Framer Motion
- + Leaflet maps



BACKEND

FastAPI · Python 3.11

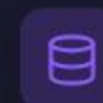
- + async / SSE streaming
- + Uvicorn workers
- + Pydantic v2
- + SQLAlchemy ORM



AI / ORCHESTRATION

LangGraph · Gemini

- + 9-agent DAG runtime
- + Google Gemini API
- + Structured prompts
- + JSON-mode outputs



DATA

PostgreSQL · PostGIS

- + Supabase managed
- + Realtime subscribe
- + Supabase Storage
- + Row-level security



AUTH · SECURITY

Supabase Auth

- + JWT sessions
- + 7 role tiers
- + RLS policies
- + Immutable audit log



DEPLOYMENT

Docker · nginx

- + Docker Compose
- + Multi-stage builds
- + nginx reverse proxy
- + Kubernetes-ready



MAPS · GEOSPATIAL

Leaflet · CARTO

- + Dark tile theme
- + CircleMarker heatmap
- + Draggable MapPicker
- + OpenStreetMap



QUALITY · CI

Playwright · pytest

- + 19-route E2E suite
- + Console-error gate
- + pytest fixtures
- + GitHub Actions

◆ THANK YOU · Q&A

*Let's build
smarter cities.*

TEAM

Vijay Barhate
Vedant Saikhede

SUBMISSION

Lenovo LEAP · AICTE
Hackathon 2.0 · 2026

FOCUS AREA

Smart Cities
& Public Services